

Homework Assignment: Logistic Regression

1 Learning objectives

- Understand odds and odds ratio
- Build logistic regression models
- Interpret coefficients

2 Solutions

2.1 Problem 1: Conceptual Understanding of Logistic Regression

2.1.1 Part A

When $X = 0$,

$$\text{log-odds} = -1.5 + 0.7(0) = -1.5$$

Therefore,

$$\text{odds} = e^{-1.5} \approx 0.223$$

The probability is:

$$p = \frac{\text{odds}}{1 + \text{odds}} = \frac{0.223}{1 + 0.223} \approx 0.182$$

2.1.2 Part B

The odds ratio associated with a one-unit increase is:

$$\text{odds ratio} = e^{\theta_1} = e^{0.7} \approx 2.014$$

Interpretation: Holding everything else constant, a one-unit increase in biomarker X is associated with approximately 2.01 times the odds of disease, or approximately a 101% increase in the odds of disease.

2.1.3 Part C

For $X = 1$:

$$z = -1.5 + 0.7(1) = -0.8,$$

so

$$p = \frac{1}{1 + e^{0.8}} \approx 0.310$$

For $X = 2$:

$$z = -1.5 + 0.7(2) = -0.1$$

so

$$p = \frac{1}{1 + e^{0.1}} \approx 0.475$$

Thus:

X	Log-odds	Probability
0	-1.5	0.182
1	-0.8	0.310
2	-0.1	0.475

The probability increases are:

$$0.310 - 0.182 = 0.128$$

and

$$0.475 - 0.310 = 0.165$$

Therefore, a one-unit increase in X does not produce a constant increase in probability.

The coefficient produces a constant change of 0.7 in log-odds, or a constant multiplication of the odds by $e^{0.7}$, but the corresponding change in probability depends on the starting probability.

2.1.4 Part D

With a threshold of 0.5, the decision boundary occurs where

$$p = 0.5$$

For the sigmoid,

$$p = 0.5 \iff z = 0$$

Therefore,

$$-1.5 + 0.7X = 0$$

and

$$X = \frac{1.5}{0.7} \approx 2.143$$

Thus

$$\hat{Y} = \begin{cases} 1 & X \geq 2.143 \\ 0 & X < 2.143 \end{cases}$$

So the model predicts disease when $X \geq 2.143$ using the 0.5 threshold.

2.1.5 Part E

The statement:

“Every one-unit increase in the biomarker increases the probability of disease by 70%.”

is **incorrect**.

The coefficient 0.7 represents a 0.7-unit increase in the log-odds of disease for every one-unit increase in X . Equivalently,

$$e^{0.7} \approx 2.014$$

so a one-unit increase in X multiplies the odds of disease by approximately 2.01.

It does not imply a 70% increase in probability. Probability changes are nonlinear and depend on the individual's starting probability.

2.1.6 Part F

If X were standardized using `StandardScaler`, a one-unit change in the transformed X corresponds to a one-standard-deviation increase in the original biomarker.

The appropriate interpretation would therefore be:

Holding all other predictors constant, a one-standard-deviation increase in biomarker X is associated with approximately $e^{0.7} = 2.01$ times the odds of disease.